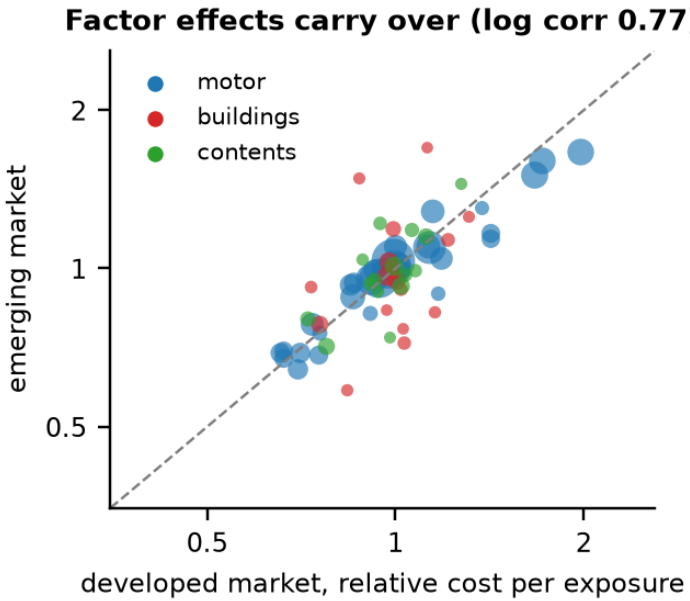
# Pricing across markets with an AI-assisted workflow

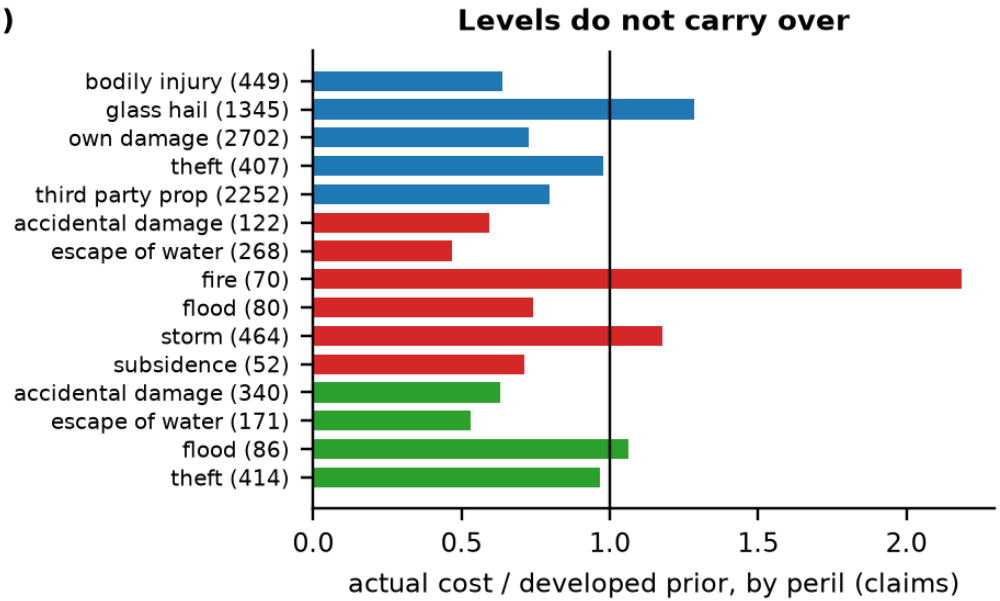
Team SU² (Kaggle team HS) · IAA AI-Assisted Actuarial Hackathon 2026

**0.13755** Part 1 private $D^2$, rank 8 of 54

**0.17556** Part 2 public $D^2$, emerging market

# What changed in the emerging market?



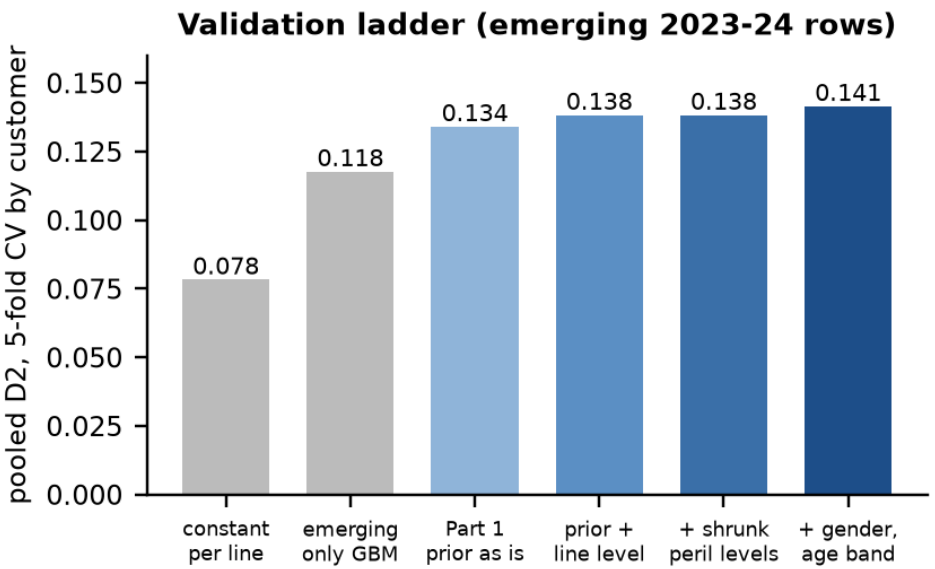

- Effects of rating factors carry over; levels do not
- Two years of data; 76% of the 2025 rows are new business

# Approach: keep relativities, re-learn levels

1. **Prior**: shallow boosting per line and peril, 4 developed years, shared fields
2. **Peril levels** from the emerging data, shrunk to the line level
3. **Corrections** only if proven: robust score test, ridge GLM, nested CV
4. **Total** restored; **new-business level** calibrated (slide 5)

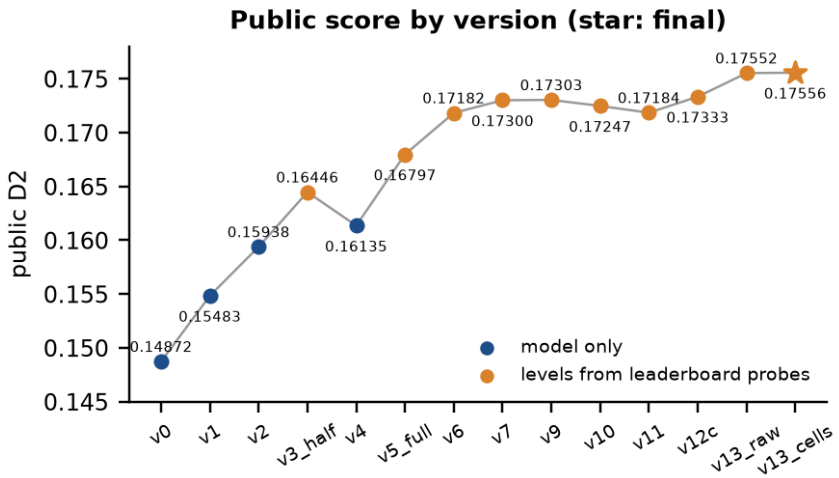
Survived: males x1.39 on at-fault motor perils; buildings age 35-49 x0.67

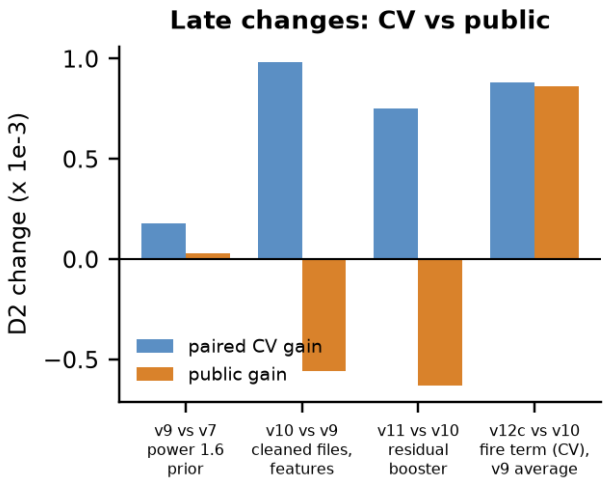
# Validation ladder and guardrails




- CV grouped by customer
- 22 automated tests
- Bugs found, now tests: column order (0.036 vs 0.134), double level (+12%), timber coded T/W

# New business: measured on the leaderboard




- Public score is separable by segment: one probe per segment gives its best multiplier
- Predicted 0.16433, observed 0.16446

# AI-assisted workflow, verified

| AI did | Humans controlled |
|---|---|
| EDA, drift, models, 40+ experiments | Adoption rule, fixed folds, nested CV |
| 22 tests from observed failures | Tests before every submission |
| Report and log | Numbers copied from script output; team picks the final |

Planted-effect and null tests check the AI's own code; rejected ideas stay in the log

# Limits and next

- New business: largest uncertainty (broker vs direct unresolved)
- Small perils shrunk hard; 2025 weather shift not modelled
- Next: collect broker-portfolio claims early